

Qualification Pack



Electric Vehicle Service Assistant

Electives: Four Wheeler Electric Vehicle Service Assistant/ Two/Three wheeler Electric Vehicle Service Assistant/ Electric Truck/Bus Service Assistant

QP Code: ASC/Q1435

Version: 2.0

NSQF Level: 2.5

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Qualification Pack

Contents

ASC/Q1435: Electric Vehicle Service Assistant	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Elective 1: Four Wheeler Electric Vehicle Service Assistant</i>	3
<i>Elective 2: Two/Three wheeler Electric Vehicle Service Assistant</i>	3
<i>Elective 3: Electric Truck/Bus Service Assistant</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
ASC/N9801: Organize work and resources (Service)	5
ASC/N1454: Assist in service, maintenance and repair of electric vehicle.	11
DGT/VSQ/N0101: Employability Skills (30 Hours)	17
ASC/N1455: Automotive Detailer Technician	23
ASC/N1456: Perform routine service, repair and maintenance of bicycle	30
ASC/N1457: Carry out routine service or minor repairs on electric bicycles and assist in diagnosis ...	36
Assessment Guidelines and Weightage	42
<i>Assessment Guidelines</i>	42
<i>Assessment Weightage</i>	43
Acronyms	45
Glossary	46

Qualification Pack

ASC/Q1435: Electric Vehicle Service Assistant

Brief Job Description

An Electric Vehicle Service Assistant is responsible for assisting the service technician in repair, routine servicing and maintenance (including electrical and mechanical aggregates) of Electric and Hybrid vehicles.

Personal Attributes

An individual in this job must have good communication and interpersonal skills. The person should be patient, organised, team-oriented, customer centric and have the ability to work for long hours in adverse conditions. The individual should be a keen observer and have an eye for detail and quality.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ASC/N9801: Organize work and resources \(Service\)](#)
2. [ASC/N1454: Assist in service, maintenance and repair of electric vehicle.](#)
3. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Electives (mandatory to select at least one):

Elective 1: Four Wheeler Electric Vehicle Service Assistant

This NOS unit is about assisting the technician and performing all tasks related to service, minor repair and diagnosis of the four wheeler electric or hybrid vehicles.

1. [ASC/N1455: Automotive Detailer Technician](#)

Elective 2: Two/Three wheeler Electric Vehicle Service Assistant

This NOS unit is about assisting the technician and performing all tasks related to service, minor repair and diagnosis on electric two/three wheeler electric vehicle.

1. [ASC/N1456: Perform routine service, repair and maintenance of bicycle](#)

Elective 3: Electric Truck/Bus Service Assistant

This NOS unit is about assisting the technician and performing all tasks related to service, minor repair and diagnosis on electric truck/bus.

1. [ASC/N1457: Carry out routine service or minor repairs on electric bicycles and assist in diagnosis](#)

Qualification Pack

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
Country	India
NSQF Level	2.5
Credits	19
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3115.0602
Minimum Educational Qualification & Experience	8th Class with 1 Year of experience of relevant experience OR 8th Class (+ 2 years ITI) OR Certificate-NSQF (Automotive Washer Level 1) with 2 Years of experience OR 10th Class
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	Driving License and Basic Computer Skills
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	28/07/2025
NSQC Approval Date	28/07/2022
Version	2.0
Reference code on NQR	QG-2.5-AU-00698-2023-V1.1-ASDC
NQR Version	1.1

Qualification Pack

ASC/N9801: Organize work and resources (Service)

Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising use of resources

Scope

The scope covers the following :

- Maintain safe and secure working environment
- Perform work as per quality standards
- Health and hygiene
- Material/energy conservation practices
- Effective waste management practices

Elements and Performance Criteria

Maintain safe and secure working environment

To be competent, the user/individual on the job must be able to:

- PC1.** organise work as per organisation's current health, safety and security policies and procedures
- PC2.** report any identified breaches in health, safety, and security policies and procedures to the designated person
- PC3.** identify the risks and hazards associated with work activities, their causes and prevention

Perform work as per quality standards

To be competent, the user/individual on the job must be able to:

- PC4.** ensure work area is clean and tidy
- PC5.** ensure that work is accomplished as per the requirements within the specified timeline
- PC6.** ensure team goals are given preference over individual goals

Health and hygiene

To be competent, the user/individual on the job must be able to:

- PC7.** sanitize workstation and equipment regularly
- PC8.** clean hands with soap, alcohol-based sanitizer regularly
- PC9.** avoid contact with ill people and self-isolate in a similar situation
- PC10.** wear and dispose PPEs regularly and appropriately
- PC11.** report advanced hygiene and sanitation issues to appropriate authority
- PC12.** follow stress and anxiety management techniques

Material/energy conservation practices

To be competent, the user/individual on the job must be able to:

- PC13.** identify ways to optimise usage of material in various tasks/activities/processes
- PC14.** use resources, including water, in a responsible manner

Qualification Pack

- PC15.** check for spills/leakages in various tasks/activities/processes
- PC16.** plug spills/leakages and escalate to appropriate authority if unable to rectify
- PC17.** carry out routine cleaning of tools, machines and equipment
- PC18.** check if the equipment/machine is functioning normally before commencing work and rectify wherever required
- PC19.** report malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment
- PC20.** ensure electrical equipment and appliances are properly connected and turned off when not in use

Effective waste management practices

To be competent, the user/individual on the job must be able to:

- PC21.** identify recyclable and non-recyclable, and hazardous waste generated
- PC22.** segregate waste into different categories
- PC23.** dispose non-recyclable waste appropriately
- PC24.** deposit recyclable and reusable material at identified location
- PC25.** follow processes specified for disposal of hazardous waste

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organisation procedures for health, safety and security, and individual role and responsibilities in this context
- KU2.** the organisations emergency procedures for different emergency situations and the importance of following the same
- KU3.** evacuation procedures for workers and visitors
- KU4.** how and when to report hazards as well as the limits of responsibility for dealing with hazards
- KU5.** potential hazards, risks and threats based on the nature of work
- KU6.** the implications of own work on the schedule and work of others
- KU7.** efficient utilisation of material and water
- KU8.** basics of electricity and prevalent energy efficient devices
- KU9.** ways to recognise common electrical problems
- KU10.** common practices of conserving electricity
- KU11.** common sources of pollution and ways to minimize it
- KU12.** categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU13.** usage of different colours of dustbins
- KU14.** waste management and methods of waste disposal
- KU15.** significance of greening
- KU16.** organisation's policies to maintain personal health and hygiene at workplace

Generic Skills (GS)

Qualification Pack

User/individual on the job needs to know how to:

- GS1.** read instructions/guidelines/standard operating procedures
- GS2.** complete statutory documents relevant to safety and hygiene
- GS3.** modify work practices to improve them
- GS4.** ask for clarifications from superior about the job requirement
- GS5.** work with supervisors/team members to carry out work related tasks
- GS6.** complete tasks efficiently and accurately within stipulated time
- GS7.** inform/report to concerned person in case of any problem
- GS8.** make timely decisions for efficient utilization of resources
- GS9.** write in at least one language and complete written work with attention to detail
- GS10.** record data on waste disposal at workplace
- GS11.** be punctual, utilize time and manage workload efficiently
- GS12.** evaluate strategies to maintain, enhance or reduce the intensity of heightened emotional response

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain safe and secure working environment</i>	8	4	-	3
PC1. organise work as per organisation's current health, safety and security policies and procedures	-	2	-	1
PC2. report any identified breaches in health, safety, and security policies and procedures to the designated person	3	1	-	-
PC3. identify the risks and hazards associated with work activities, their causes and prevention	5	1	-	2
<i>Perform work as per quality standards</i>	12	8	-	6
PC4. ensure work area is clean and tidy	4	2	-	-
PC5. ensure that work is accomplished as per the requirements within the specified timeline	6	4	-	2
PC6. ensure team goals are given preference over individual goals	2	2	-	4
<i>Health and hygiene</i>	12	8	-	5
PC7. sanitize workstation and equipment regularly	2	2	-	2
PC8. clean hands with soap, alcohol-based sanitizer regularly	2	1	-	-
PC9. avoid contact with ill people and self-isolate in a similar situation	2	1	-	-
PC10. wear and dispose PPEs regularly and appropriately	2	2	-	1
PC11. report advanced hygiene and sanitation issues to appropriate authority	2	2	-	2
PC12. follow stress and anxiety management techniques	2	-	-	-
<i>Material/energy conservation practices</i>	10	4	-	3
PC13. identify ways to optimise usage of material in various tasks/activities/processes	2	-	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. use resources, including water, in a responsible manner	2	-	-	-
PC15. check for spills/leakages in various tasks/activities/processes	-	1	-	-
PC16. plug spills/leakages and escalate to appropriate authority if unable to rectify	-	1	-	1
PC17. carry out routine cleaning of tools, machines and equipment	2	-	-	-
PC18. check if the equipment/machine is functioning normally before commencing work and rectify wherever required	-	1	-	1
PC19. report malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment	2	1	-	-
PC20. ensure electrical equipment and appliances are properly connected and turned off when not in use	2	-	-	-
<i>Effective waste management practices</i>	8	6	-	3
PC21. identify recyclable and non-recyclable, and hazardous waste generated	2	-	-	1
PC22. segregate waste into different categories	-	2	-	-
PC23. dispose non-recyclable waste appropriately	2	2	-	1
PC24. deposit recyclable and reusable material at identified location	2	1	-	-
PC25. follow processes specified for disposal of hazardous waste	2	1	-	1
NOS Total	50	30	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9801
NOS Name	Organize work and resources (Service)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

ASC/N1454: Assist in service, maintenance and repair of electric vehicle.

Description

This NOS unit is about assisting the technician in repair, routine servicing and maintenance of electric vehicle.

Scope

The scope covers the following :

- Assist in preparation for service, maintenance and repair of the vehicle
- Assist in routine service and minor repairs
- Perform post service/repair/diagnostic activities

Elements and Performance Criteria

Assist in preparation for service, maintenance and repair of the vehicle

To be competent, the user/individual on the job must be able to:

- PC1.** obtain the instructions from service technician related to work to be carried out on the electric vehicle
- PC2.** identify the auto components related to the various aggregates in the electric vehicle
- PC3.** ensure no HV (High Voltage) activity is being conducted around workstation prior to commencement of work
- PC4.** collect workshop tools/measuring devices/equipment required to carry out job on electric vehicle and check their condition/calibration as per the instructions of service technician
- PC5.** prepare the work area by cleaning and placing the tools/equipment in an organised manner
- PC6.** assist in positioning the electrical vehicle on a suitable platform for service, maintenance and repair work and safeguard the working area during electrical work
- PC7.** wear PPE according to nature of job to be performed on electric vehicle
- PC8.** assist service technician in visual inspection of the electric vehicle to identify defects in HV components and indirect faults in electrical/electronic aggregate due to other system/component
- PC9.** assist service technician in assessing the mechanical aggregates such as steering gear, suspension, axle, brakes, etc. of the electric vehicle for any external impact/bend/leak/incorrect level/wear & tear
- PC10.** report the malfunctions/repairs in the electric vehicle beyond own scope to the concerned person

Assist in routine service and minor repairs

To be competent, the user/individual on the job must be able to:

- PC11.** work carefully to ensure no damage is caused to any aggregate/component of electric vehicle
- PC12.** assist in dismantling or removing parts relevant to various mechanical aggregates of electric vehicle and place them securely as specified by OEM

Qualification Pack

- PC13.** clean and condition dismantled mechanical and electrical components of electric vehicle prior to assembly
- PC14.** assist in performing routine service/ maintenance (inspection/ correction/ adjustment/ cleaning/ lubrication) of electric vehicle/ aggregate
- PC15.** perform routine service/ maintenance (change/ replenish/top-up - lubricants, fluids, coolant, filters etc.) and minor repair/ replacement (wheels, wiper, mud flap, brake pad, brake shoes, battery, wheel balancing etc.) in electric vehicle/aggregate under supervision of senior
- PC16.** maintain the documentation related to inspection, servicing and minor repair performed on the electric vehicle

Post service/repair/diagnostic activities

To be competent, the user/individual on the job must be able to:

- PC17.** assist in checking the performance of electric vehicle/aggregate post repair
- PC18.** check and confirm that battery charge, battery water, brake oil, gear oil, engine oil etc. are filled
- PC19.** dispose of materials such as old batteries, scrap of failed parts/aggregates as per organization's policies
- PC20.** return leftover consumable/parts, tools/equipment, and report if any malfunctions are observed to the person concerned

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the automotive industry in India, workshop structure and role and responsibilities of different people in the workshop
- KU2.** SOP for receiving vehicles, opening job card, allocation of work, invoicing, vehicle delivery, handling complaints, etc.
- KU3.** different components/aggregates as well as auto component manufacturer's specifications of the electric vehicle
- KU4.** basic technology used in functioning of various systems and components of the electric vehicle including electrical machines and devices such as: DC/DC converters, AC motor, DC motor, charging systems etc.
- KU5.** interconnection of systems with each other and effect of one system on other system
- KU6.**
 - fundamental terms, laws and principles of electricity used in electric vehicles such as:
 - principles of storing electrical voltage, ohms law, voltage, current (AC/DC/HV), resistance,
 - power, capacitance, electrostatics, magnetic, inductance, discrete electronic components, and radio frequency (automotive digital computers, automotive communication protocols such as CAN, LIN, MOST, etc.)
- KU7.** use of relevant measuring device/equipment and interpretation of all relevant mathematical calculations
- KU8.** various electrical and electronic signals such as electrical inputs, outputs, voltage, pulsewidth modulation, digital signal (including infra-red and fiber optics) etc.
- KU9.** symbols, units and terms used in wiring diagrams associated with electrical/electric systems/components of the electric vehicle
- KU10.** how to use a computer, online application and OEM technical information/assistance portals

Qualification Pack

- KU11.** various sources of information available for assessing service and repair requirements of the electric vehicle including diagnostic displays, visual inspections, test drives, vehicle/equipment manufacturer specifications, and tolerance limits of components
- KU12.** standard schedules and checklists recommended by the OEM/auto component manufacturer for servicing of electric vehicle
- KU13.** typical symptoms of common faults and failures of mechanical, electrical and electronic systems of the electric vehicle
- KU14.** safety, health and environmental policies and regulations for the workplace as well as for automotive trade in general
- KU15.** Standard Operating Procedures (SOPs) of the organization/ dealership for inspection and diagnosis of faults in the electric vehicle as prescribed by the OEM/components manufacturer
- KU16.** how to work on the HV systems which do not require isolation, troubleshooting and replacing parts on the active HV system
- KU17.** SOP recommended by OEM for using tools/equipment for diagnosis or troubleshooting such as special service tools, measuring instrument, volt meters, ammeters, ohmmeters, battery tester, dedicated and computer based diagnostic equipment, oscilloscopes, etc.
- KU18.** various workshop tools/measuring devices/equipment required to carry out job on electric vehicle and their common errors or defects
- KU19.** documentation requirements for each procedure carried out as part of roles and responsibilities as specified by OEM/ auto component manufacturer
- KU20.** organizational/professional code of ethics and standards of practice
- KU21.** electrical hazards, protective measures and first aid: in case of electric shock, electrical arc in public grid or in an electrical vehicle, impact of electric current/arc, secondary accidents
- KU22.** five safety rules for electrical work on HV systems before starting the work i.e. isolate, safeguard reconnection, verify the non-live state, earth or short-circuit and shroud or safeguard adjacent live parts
- KU23.** safety requirements recommended by the OEM for equipment/electric vehicle components during diagnosis, troubleshooting and root cause analysis on various aggregates
- KU24.** legal regulations that need to be taken into account for handling electric vehicles in the workshop
- KU25.** Occupational Safety and Health (OSH) measures required for working on electric vehicle

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret workplace related documentation
- GS2.** interpret the needs of customers by understanding the key issues
- GS3.** communicate using terms, names, grades and other nomenclature pertaining to the automotive trade
- GS4.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS5.** identify potential workplace problem and take suitable action
- GS6.** write in English/regional language

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Assist in preparation for service, maintenance and repair of the vehicle</i>	10	15	-	7
PC1. obtain the instructions from service technician related to work to be carried out on the electric vehicle	-	-	-	-
PC2. identify the auto components related to the various aggregates in the electric vehicle	-	-	-	-
PC3. ensure no HV (High Voltage) activity is being conducted around workstation prior to commencement of work	-	-	-	-
PC4. collect workshop tools/measuring devices/equipment required to carry out job on electric vehicle and check their condition/calibration as per the instructions of service technician	-	-	-	-
PC5. prepare the work area by cleaning and placing the tools/equipment in an organised manner	-	-	-	-
PC6. assist in positioning the electrical vehicle on a suitable platform for service, maintenance and repair work and safeguard the working area during electrical work	-	-	-	-
PC7. wear PPE according to nature of job to be performed on electric vehicle	-	-	-	-
PC8. assist service technician in visual inspection of the electric vehicle to identify defects in HV components and indirect faults in electrical/electronic aggregate due to other system/component	-	-	-	-
PC9. assist service technician in assessing the mechanical aggregates such as steering gear, suspension, axle, brakes, etc. of the electric vehicle for any external impact/bend/leak/incorrect level/wear & tear	-	-	-	-
PC10. report the malfunctions/repairs in the electric vehicle beyond own scope to the concerned person	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Assist in routine service and minor repairs</i>	10	17	-	7
PC11. work carefully to ensure no damage is caused to any aggregate/component of electric vehicle	-	-	-	-
PC12. assist in dismantling or removing parts relevant to various mechanical aggregates of electric vehicle and place them securely as specified by OEM	-	-	-	-
PC13. clean and condition dismantled mechanical and electrical components of electric vehicle prior to assembly	-	-	-	-
PC14. assist in performing routine service/ maintenance (inspection/ correction/ adjustment/ cleaning/ lubrication) of electric vehicle/ aggregate	-	-	-	-
PC15. perform routine service/ maintenance (change/ replenish/top-up - lubricants, fluids, coolant, filters etc.) and minor repair/ replacement (wheels, wiper, mud flap, brake pad, brake shoes, battery, wheel balancing etc.) in electric vehicle/aggregate under supervision of senior	-	-	-	-
PC16. maintain the documentation related to inspection, servicing and minor repair performed on the electric vehicle	-	-	-	-
<i>Post service/repair/diagnostic activities</i>	10	18	-	6
PC17. assist in checking the performance of electric vehicle/aggregate post repair	-	-	-	-
PC18. check and confirm that battery charge, battery water, brake oil, gear oil, engine oil etc. are filled	-	-	-	-
PC19. dispose of materials such as old batteries, scrap of failed parts/aggregates as per organization's policies	-	-	-	-
PC20. return leftover consumable/parts, tools/equipment, and report if any malfunctions are observed to the person concerned	-	-	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1454
NOS Name	Assist in service, maintenance and repair of electric vehicle.
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	4
Credits	7
Version	1.0
Last Reviewed Date	28/07/2022
Next Review Date	28/07/2025
NSQC Clearance Date	28/07/2022

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

ASC/N1455: Automotive Detailer Technician

Description

This NOS unit is about performing and supporting detailer assistant in all tasks related to vehicle detailing such as steam cleaning of vehicle interior and exterior, correction of paint defects and polishing of vehicle parts. This is also about applying protective coating like ceramic / nano coating / Paint Protection Film (PPF) etc. on the vehicle for body protection.

Scope

The scope covers the following :

- Prepare for vehicle detailing work
- Perform steam cleaning of vehicle
- Perform polishing and correction of paint defects
- Perform post-vehicle detailing activities

Elements and Performance Criteria

Prepare for vehicle detailing work

To be competent, the user/individual on the job must be able to:

- PC1.** determine work to be done and work requirements by obtaining required information from the job card and instructions received from supervisor
- PC2.** identify appropriate material, tools & equipment such as cleaning agent, polish, foam, brushes, mops, microfiber cloth, solvent sprayer, vacuum cleaner, buffers, hoses, steamers, protective coatings etc. required for work
- PC3.** check material, tools & equipment for any issues and required quality standards before use
- PC4.** report the defects/malfunctions if any, in the tools and equipment to the concerned person for rectification
- PC5.** inspect the vehicle interior & exterior for cleaning and detailing work
- PC6.** support the detailer assistant in removing detachable items, seats etc. and place them securely at specified location
- PC7.** support the detailer assistant in mixing cleaning solutions, abrasive compositions or other compounds as per requirement and SOP

Perform steam cleaning of vehicle

To be competent, the user/individual on the job must be able to:

- PC8.** follow safety precautions to avoid any hazard and accident during work
- PC9.** carry out steam cleaning of the vehicle interior & exterior components with the assistance of detailer assistant by following SOP/WI
- PC10.** ensure that all the segments and surfaces of the vehicle such as carpet, seats, trims, headlining, consoles, grill, vents, body panels, wheels, etc. are cleaned properly as per the SOP/WI
- PC11.** clean all exposed glass like windshield, side windows, side mirrors, rear window, headlight, fog light, taillight lenses etc. by using appropriate glass cleaner and microfiber cloths

Qualification Pack

Perform polishing and correction of paint defects

To be competent, the user/individual on the job must be able to:

- PC12.** perform polishing with appropriate abrasive compound and buffers on painted components of vehicle as per SOP/WI
- PC13.** identify minor paint imperfections such as scratches, swirl marks etc. and any remaining contamination on the vehicle parts/panels
- PC14.** ensure that detailer assistant covered the exterior parts/fittings adjacent to the painted panels with masking tape or paper to prevent their surface finish
- PC15.** remove the minor paint imperfections by applying appropriate paint compounds and pads on the panels with the assistance of detailer assistant by following SOP/WI
- PC16.** use sandpaper with a specific set of grids to sand a particular area for removal of the defects
- PC17.** remove any contaminants, overspray, or residue left by using clay bar with the assistance of detailer assistant after the cleaning and polishing
- PC18.** apply paint sealer/wax on the vehicle parts to give final glossy shine on the painted panels/parts of vehicle by following SOP/WI
- PC19.** apply protective coating like ceramic / nano coating / Paint Protection Film (PPF) etc. on the vehicle to protect the paint of vehicle by following SOP/WI

Perform post vehicle detailing activities

To be competent, the user/individual on the job must be able to:

- PC20.** inspect surface finish of vehicle parts/panels after vehicle detailing activities and report to supervisor in case of any discrepancy
- PC21.** ensure that all the cleaning and detailing activities are completed before releasing the vehicle
- PC22.** dispose off materials such as debris, old masking paper, empty cans/tube, etc. as per organization's policies
- PC23.** return leftover consumable/parts, tools and equipment after completion of work
- PC24.** record the details of various treatment and washing activities done on the vehicle in job card as per organizational procedures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant legislation, standards, policies, and procedures followed in the organization
- KU2.** SOPs regarding receiving vehicles, opening job card, allocation of work, invoicing, vehicle delivery, handling complaints etc.
- KU3.** identification, nomenclature, manufacturer's specifications of various consumable/material (abrasives, solvent, masking paper, cloth etc.
- KU4.** various interior & exterior segments such as seat (fabric/PU/leathered), dash panel, engine compartment, bonnet, dickey etc. and detachable items such as foot-mats, mobile charger etc. of the vehicle
- KU5.** Standard Operating Procedures of the dealership for vehicle washing, cleaning and detailing
- KU6.** SOP recommended by manufacturer for using cleaning agents, material, tools and equipment during work

Qualification Pack

- KU7.** different types of grime such as scuff marks, soil, dust, oil, grease, dried food and other stains
- KU8.** methods like steam cleaning, brushing, water/air spray, use of chemicals, manual or machine assisted cleaning for cleaning various grime
- KU9.** application method of various polish & abrasive, on different surfaces such as dashboard, seats, door trim, tyre, bumper, front grill etc. using buffer/polisher
- KU10.** methods of using different squeegees, mops, cloth, vacuum cleaner, steamer, buffing machine, buffing pads, etc.
- KU11.** various types of paint protection coatings like ceramic / nano coating / Paint Protection Film (PPF) and how to apply them
- KU12.** safety, health and environmental policies and regulations of the workplace as well as for automotive trade in general (e.g. safe practices while working in pits/under vehicles)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret workplace related documentation
- GS2.** interpret the needs of customers by understanding the key issues
- GS3.** communicate using terms, names, grades and other nomenclature pertaining to the automotive trade
- GS4.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS5.** identify potential workplace problem and take suitable action
- GS6.** write in English/regional language

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for vehicle detailing work</i>	8	10	-	4
PC1. determine work to be done and work requirements by obtaining required information from the job card and instructions received from supervisor	2	1	-	-
PC2. identify appropriate material, tools & equipment such as cleaning agent, polish, foam, brushes, mops, microfiber cloth, solvent sprayer, vacuum cleaner, buffers, hoses, steamers, protective coatings etc. required for work	2	2	-	1
PC3. check material, tools & equipment for any issues and required quality standards before use	2	1	-	1
PC4. report the defects/malfunctions if any, in the tools and equipment to the concerned person for rectification	-	1	-	1
PC5. inspect the vehicle interior & exterior for cleaning and detailing work	-	2	-	1
PC6. support the detailer assistant in removing detachable items, seats etc. and place them securely at specified location	-	1	-	-
PC7. support the detailer assistant in mixing cleaning solutions, abrasive compositions or other compounds as per requirement and SOP	2	2	-	-
<i>Perform steam cleaning of vehicle</i>	6	14	-	6
PC8. follow safety precautions to avoid any hazard and accident during work	2	5	-	2
PC9. carry out steam cleaning of the vehicle interior & exterior components with the assistance of detailer assistant by following SOP/WI	2	3	-	2
PC10. ensure that all the segments and surfaces of the vehicle such as carpet, seats, trims, headlining, consoles, grill, vents, body panels, wheels, etc. are cleaned properly as per the SOP/WI	1	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. clean all exposed glass like windshield, side windows, side mirrors, rear window, headlight, fog light, taillight lenses etc. by using appropriate glass cleaner and microfiber cloths	1	3	-	1
<i>Perform polishing and correction of paint defects</i>	11	18	-	6
PC12. perform polishing with appropriate abrasive compound and buffers on painted components of vehicle as per SOP/WI	2	3	-	-
PC13. identify minor paint imperfections such as scratches, swirl marks etc. and any remaining contamination on the vehicle parts/panels	2	2	-	1
PC14. ensure that detailer assistant covered the exterior parts/fittings adjacent to the painted panels with masking tape or paper to prevent their surface finish	1	2	-	-
PC15. remove the minor paint imperfections by applying appropriate paint compounds and pads on the panels with the assistance of detailer assistant by following SOP/WI	2	3	-	1
PC16. use sandpaper with a specific set of grids to sand a particular area for removal of the defects	1	2	-	1
PC17. remove any contaminants, overspray, or residue left by using clay bar with the assistance of detailer assistant after the cleaning and polishing	1	2	-	1
PC18. apply paint sealer/wax on the vehicle parts to give final glossy shine on the painted panels/parts of vehicle by following SOP/WI	1	2	-	1
PC19. apply protective coating like ceramic / nano coating / Paint Protection Film (PPF) etc. on the vehicle to protect the paint of vehicle by following SOP/WI	1	2	-	1
<i>Perform post vehicle detailing activities</i>	5	8	-	4
PC20. inspect surface finish of vehicle parts/panels after vehicle detailing activities and report to supervisor in case of any discrepancy	1	2	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. ensure that all the cleaning and detailing activities are completed before releasing the vehicle	1	2	-	-
PC22. dispose off materials such as debris, old masking paper, empty cans/tube, etc. as per organization's policies	2	2	-	1
PC23. return leftover consumable/parts, tools and equipment after completion of work	1	2	-	1
PC24. record the details of various treatment and washing activities done on the vehicle in job card as per organizational procedures	-	-	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1455
NOS Name	Automotive Detailer Technician
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	28/07/2022
Next Review Date	28/07/2025
NSQC Clearance Date	28/07/2022

Qualification Pack

ASC/N1456: Perform routine service, repair and maintenance of bicycle

Description

This NOS unit is about performing routine service, repair and maintenance activities of various aggregates/systems of a bicycle.

Scope

The scope covers the following :

- Prepare for work
- Perform routine service, repair or maintenance of bicycle
- Perform post service/repair activities

Elements and Performance Criteria

Prepare for work

To be competent, the user/individual on the job must be able to:

- PC1.** interact with customer to understand their requirement of routine service, minor repairs, maintenance and warranty replacement of bicycle
- PC2.** collect tools/measuring devices/equipment required for the job and check their condition/calibration
- PC3.** prepare the bicycle for service, repair and maintenance as per SOP
- PC4.** wear PPE according to nature of job to be performed
- PC5.** conduct visual inspection of the bicycle to identify defects in systems/components
- PC6.** assess mechanical aggregates of the bicycle for any external impact/bend/incorrect level/wear & tear/fault
- PC7.** provide cost and time estimate to the customer for the service and repair work
- PC8.** get the customer's consent and proceed to the work

Perform routine service, repair or maintenance of bicycle

To be competent, the user/individual on the job must be able to:

- PC9.** perform routine service/maintenance of various parts and aggregates including drive chain ring, steering system, brakes, suspension, pedals etc.
- PC10.** collect the correct spare parts and appropriate grade of lubricants, coolant, oils and grease for routine maintenance and service
- PC11.** inspect the components requiring adjustment or replacement due to continuous wear and tear such as bell, tires, reflectors, brake pads, chain ring, cassettes and freewheels etc.
- PC12.** check that tire pressure is as per the manufacturer's specifications by using pressure gauge, refill the air (if required)
- PC13.** check that brake is tightly gripping the wheel when the brake levers are squeezed, adjust the brake pads as per the requirement
- PC14.** adjust spoke tension and replace any damaged spokes/ true up the wheel
- PC15.** check the socket and bolt for rust and worn out threads, replace the sockets if required

Qualification Pack

- PC16.** clean and condition dismantled parts/components, prior to reassembly
- PC17.** overhaul hub, steering system (handlebars, stem and headset), headrest, pedals, crank arm and bottom bracket as per the need
- PC18.** perform lubrication or greasing of chain, freewheel or cassette and other parts as per the requirement
- PC19.** carry out re-assembling of bicycle parts/ aggregates as per SOP
- PC20.** adjust the saddle fore, aft and height as per the requirement
- PC21.** seek assistance from the specialist in case of structural repairs

Perform post service/repair activities

To be competent, the user/individual on the job must be able to:

- PC22.** check the performance of bicycle/aggregate post repair before releasing it to customer
- PC23.** dispose-off materials such as waste oil, scrap of failed parts/aggregates, as per environmental policies
- PC24.** perform scheduled checks, calibration and timely repairs for workshop tools and equipment after completion of work

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the automotive industry in India, workshop structure and role and responsibilities of different people in the workshop
- KU2.** various brands of bicycles available in the market
- KU3.** types of bicycles
- KU4.** different components/aggregates and systems i.e. drive train, steering system, suspension system, brakes, wheels, structural members as well as component manufacturer's specifications of the bicycle
- KU5.** basic technology used in and functioning of various systems and components of the bicycle
- KU6.**
 - different types of braking system
 - Types of brakes: Rod actuated brakes, caliper brake, cantilever brake, V brake safety features provided in a bicycle Band brake, Drum or hub brake, Backpedal or coaster brake, Disc brake and Roller brake
- KU7.**
 - safety features provided in a bicycle
 - Safety Features: Bell, reflector, brake, chain guard etc.
- KU8.** standard schedules and checklists recommended by the OEM/auto component manufacturer for servicing of bicycles
- KU9.** typical symptoms of common faults and failures in bicycle
- KU10.** Standard Operating Procedures (SOPs) for inspection and diagnosis of faults in a bicycle as prescribed by the OEM/components manufacturer
- KU11.** SOP recommended by OEM for using tools/equipment for diagnosis or troubleshooting
- KU12.** how to replace tire and tube
- KU13.** how to install front and rear wheels
- KU14.** how to find a puncture, patch the tube and attach the wheel
- KU15.** method of wheel truing

Qualification Pack

- KU16.** how to repair dents in the rim
- KU17.** types of head rest i.e. Threaded and thread less
- KU18.** types of suspension i.e. Coil and air suspension system
- KU19.** chain and sprocket mechanism- drive mechanism
- KU20.** how to calculate Mechanical Advantage or Velocity Ratio
- KU21.** how to remove / replace/ reassemble a chain
- KU22.** how to clean the freewheel or cassette
- KU23.** styles of hub i.e. cup and cone, sealed bearing
- KU24.** how to adjust a hub centering and tensioning of brakes
- KU25.** how to replace brake cables
- KU26.** how to angle the saddle
- KU27.** names of structural members of a frame i.e. Head tube, Top tube, Down tube, Seat tube, bottom
- KU28.** use and application of bearings
- KU29.** types of frame materials
- KU30.** Occupational Safety and Health (OSH) measures required for working on bicycles

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret workplace related documentation
- GS2.** interpret the needs of customers by understanding the key issues
- GS3.** communicate using terms, names, grades and other nomenclature pertaining to the automotive trade
- GS4.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS5.** identify potential workplace problem and take suitable action
- GS6.** read various sources of information available for assessing service and repair requirements
- GS7.** write any work-related information
- GS8.** write in English/regional language

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for work</i>	5	14	-	7
PC1. interact with customer to understand their requirement of routine service, minor repairs, maintenance and warranty replacement of bicycle	-	1	-	1
PC2. collect tools/measuring devices/equipment required for the job and check their condition/calibration	1	2	-	1
PC3. prepare the bicycle for service, repair and maintenance as per SOP	1	2	-	1
PC4. wear PPE according to nature of job to be performed	-	1	-	1
PC5. conduct visual inspection of the bicycle to identify defects in systems/components	1	1	-	1
PC6. assess mechanical aggregates of the bicycle for any external impact/bend/incorrect level/wear & tear/fault	-	3	-	1
PC7. provide cost and time estimate to the customer for the service and repair work	1	3	-	1
PC8. get the customer's consent and proceed to the work	1	1	-	-
<i>Perform routine service, repair or maintenance of bicycle</i>	21	31	-	11
PC9. perform routine service/maintenance of various parts and aggregates including drive chain ring, steering system, brakes, suspension, pedals etc.	1	2	-	-
PC10. collect the correct spare parts and appropriate grade of lubricants, coolant, oils and grease for routine maintenance and service	2	2	-	1
PC11. inspect the components requiring adjustment or replacement due to continuous wear and tear such as bell, tires, reflectors, brake pads, chain ring, cassettes and freewheels etc.	2	3	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. check that tire pressure is as per the manufacturer's specifications by using pressure gauge, refill the air (if required)	2	3	-	-
PC13. check that brake is tightly gripping the wheel when the brake levers are squeezed, adjust the brake pads as per the requirement	1	2	-	-
PC14. adjust spoke tension and replace any damaged spokes/ true up the wheel	2	3	-	1
PC15. check the socket and bolt for rust and worn out threads, replace the sockets if required	1	2	-	1
PC16. clean and condition dismantled parts/components, prior to reassembly	1	1	-	-
PC17. overhaul hub, steering system (handlebars, stem and headset), headrest, pedals, crank arm and bottom bracket as per the need	3	5	-	1
PC18. perform lubrication or greasing of chain, freewheel or cassette and other parts as per the requirement	-	2	-	-
PC19. carry out re-assembling of bicycle parts/ aggregates as per SOP	2	2	-	2
PC20. adjust the saddle fore, aft and height as per the requirement	2	2	-	2
PC21. seek assistance from the specialist in case of structural repairs	2	2	-	1
<i>Perform post service/repair activities</i>	4	5	-	2
PC22. check the performance of bicycle/aggregate post repair before releasing it to customer	1	1	-	-
PC23. dispose-off materials such as waste oil, scrap of failed parts/aggregates, as per environmental policies	1	1	-	-
PC24. perform scheduled checks, calibration and timely repairs for workshop tools and equipment after completion of work	2	3	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1456
NOS Name	Perform routine service, repair and maintenance of bicycle
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	28/07/2022
Next Review Date	28/07/2025
NSQC Clearance Date	28/07/2022

Qualification Pack

ASC/N1457: Carry out routine service or minor repairs on electric bicycles and assist in diagnosis

Description

This NOS unit is about performing all tasks related to service, minor repair and diagnosis of electric bicycles.

Scope

The scope covers the following :

- Prepare for service and repair work
- Perform routine service and minor repairs
- Assist lead technician in diagnosis or troubleshooting the faults
- Post service/repair/diagnostic activities

Elements and Performance Criteria

Prepare for service and repair work

To be competent, the user/individual on the job must be able to:

- PC1.** review the job card and understand work to be carried out
- PC2.** identify the auto components related to the various aggregates in the electric bicycle
- PC3.** collect workshop tools/measuring devices/equipment required for the job and check their condition/calibration
- PC4.** prepare the electric bicycle according to nature of job to be performed: general, mechanical or electrical job on the electric bicycle
- PC5.** wear PPE according to nature of job to be performed
- PC6.** conduct visual inspection of the electric bicycle to identify defects and indirect faults in electrical/electronic aggregate due to other system/component
- PC7.** assess mechanical aggregates such as gear shifter, handle, axles, paddles brakes, wheels, etc. of the electric bicycle for any external impact/bend/incorrect fitment/wear & tear
- PC8.** report the malfunctions/repairs in the electric bicycle beyond own scope to the concerned person

Perform routine service and minor repairs

To be competent, the user/individual on the job must be able to:

- PC9.** take precautions to avoid damage to the electric bicycle and its components while working on various aggregates
- PC10.** use appropriate tools, equipment, and consumables as per nature of job and Standard Operating Procedure (SOP) recommended by the organization
- PC11.** test electrical/electronic components performance of electric bicycle wherever applicable as per OEM SOP
- PC12.** remove parts relevant to various mechanical aggregates and place them securely as specified by OEM
- PC13.** clean and condition dismantled mechanical and electrical components prior to assembly

Qualification Pack

- PC14.** perform minor repair/replacement/calibration of mechanical system/aggregate such as brake lever free play, drive chain, braking & steering systems adjustments, etc. as per SOP
- PC15.** apply appropriate grade of oil or other lubricant on the mechanical components of electric bicycle wherever applicable as per OEM guidelines
- PC16.** maintain the documentation related to inspection, servicing and minor repair of the electric bicycle

Assist lead technician in diagnosis or troubleshooting the faults

To be competent, the user/individual on the job must be able to:

- PC17.** conduct test ride of the e-bike to assist the lead technician in assessing the service/repair requirement or calibration/adjustments, if any
- PC18.** follow lead technician instructions to choose and use the appropriate device/equipment to inspect/test electric bicycle/system/component performance to diagnose defects or faults in it
- PC19.** carry out inspection or test on electric bicycle mechanical and electrical systems according to lead technician instructions
- PC20.** interpret and compare results of diagnostic inspections/tests with vehicle specifications and regulatory requirements
- PC21.** maintain the documentation related to inspections and troubleshooting performed on the vehicle
- PC22.** report the results to lead technician and seek assistance if further tests or inspections are required to conclude the diagnosis or troubleshooting

Perform post service/repair/diagnostic activities

To be competent, the user/individual on the job must be able to:

- PC23.** check the performance of electric bicycle post repair and report to supervisor/service advisor if further inspection is required by another specialist
- PC24.** ensure completeness of tasks assigned before releasing the electric bicycle for the next procedure
- PC25.** dispose of materials such as old batteries, scrap of failed parts/aggregates as per organization's policies
- PC26.** return leftover consumable/parts, tools/equipment, and report if any malfunctions are observed to the person concerned

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different components/aggregates as well as auto component manufacturer's specifications of the electric bicycle
- KU2.** road licensing requirements for electric bicycles in India
- KU3.** basic technology used in and functioning of various systems and components of the electric bicycle such as: brakes, suspension, steering, hub drive/chain drive, etc. including electrical machines and devices used in electric vehicles such as: charging system, charger, batteries, etc.
- KU4.** interconnection of systems with each other and effect of one system on other system
- KU5.** how to use computer, on-line application and OEM technical information/assistance portals

Qualification Pack

- KU6.** various sources of information available for assessing service and repair requirements of the electric bicycle including diagnostic displays, visual inspections, test rides, vehicle/equipment manufacturer specifications, and tolerance limits of components
- KU7.** standard schedules and checklists recommended by the OEM/auto component manufacturer for servicing of electric bicycle
- KU8.** typical symptoms of common faults and failures in electric bicycle mechanical, electrical and electronic systems
- KU9.** Standard Operating Procedures (SOPs) of the organization/ dealership for inspection and diagnosis of faults in an electric bicycle as prescribed by the OEM/components manufacturer
- KU10.** different types of errors or defects in the tools/equipment
- KU11.** documentation requirements for each procedure carried out as part of roles and responsibilities as specified by OEM/ auto component manufacturer
- KU12.** safety requirements recommended by the OEM for equipment/vehicle components during diagnosis, troubleshooting and root cause analysis on various aggregates
- KU13.** legal regulations that need to be taken into account for handling an electric bicycle in the workshop
- KU14.** precautions need to follow during charging of electric bicycle battery
- KU15.** Occupational Safety and Health (OSH) measures are required for working on electric vehicle

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret workplace related documentation
- GS2.** interpret the needs of customers by understanding the key issues
- GS3.** communicate using terms, names, grades and other nomenclature pertaining to the automotive trade
- GS4.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS5.** identify potential workplace problem and take suitable action
- GS6.** read various sources of information available for assessing service and repair requirements
- GS7.** write any work-related information
- GS8.** write in English/regional language

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for service and repair work</i>	5	14	-	7
PC1. review the job card and understand work to be carried out	-	1	-	1
PC2. identify the auto components related to the various aggregates in the electric bicycle	1	2	-	1
PC3. collect workshop tools/measuring devices/equipment required for the job and check their condition/calibration	1	2	-	1
PC4. prepare the electric bicycle according to nature of job to be performed: general, mechanical or electrical job on the electric bicycle	-	1	-	1
PC5. wear PPE according to nature of job to be performed	1	1	-	1
PC6. conduct visual inspection of the electric bicycle to identify defects and indirect faults in electrical/electronic aggregate due to other system/component	-	3	-	1
PC7. assess mechanical aggregates such as gear shifter, handle, axles, paddles brakes, wheels, etc. of the electric bicycle for any external impact/bend/incorrect fitment/wear & tear	1	3	-	1
PC8. report the malfunctions/repairs in the electric bicycle beyond own scope to the concerned person	1	1	-	-
<i>Perform routine service and minor repairs</i>	12	18	-	5
PC9. take precautions to avoid damage to the electric bicycle and its components while working on various aggregates	1	2	-	-
PC10. use appropriate tools, equipment, and consumables as per nature of job and Standard Operating Procedure (SOP) recommended by the organization	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. test electrical/electronic components performance of electric bicycle wherever applicable as per OEM SOP	2	3	-	2
PC12. remove parts relevant to various mechanical aggregates and place them securely as specified by OEM	2	3	-	-
PC13. clean and condition dismantled mechanical and electrical components prior to assembly	1	2	-	-
PC14. perform minor repair/replacement/calibration of mechanical system/aggregate such as brake lever free play, drive chain, braking & steering systems adjustments, etc. as per SOP	2	3	-	1
PC15. apply appropriate grade of oil or other lubricant on the mechanical components of electric bicycle wherever applicable as per OEM guidelines	1	2	-	1
PC16. maintain the documentation related to inspection, servicing and minor repair of the electric bicycle	1	1	-	-
<i>Assist lead technician in diagnosis or troubleshooting the faults</i>	8	10	-	5
PC17. conduct test ride of the e-bike to assist the lead technician in assessing the service/repair requirement or calibration/adjustments, if any	-	2	-	-
PC18. follow lead technician instructions to choose and use the appropriate device/equipment to inspect/test electric bicycle/system/component performance to diagnose defects or faults in it	2	2	-	2
PC19. carry out inspection or test on electric bicycle mechanical and electrical systems according to lead technician instructions	2	2	-	2
PC20. interpret and compare results of diagnostic inspections/tests with vehicle specifications and regulatory requirements	2	2	-	1
PC21. maintain the documentation related to inspections and troubleshooting performed on the vehicle	1	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. report the results to lead technician and seek assistance if further tests or inspections are required to conclude the diagnosis or troubleshooting	1	1	-	-
<i>Perform post service/repair/diagnostic activities</i>	5	8	-	3
PC23. check the performance of electric bicycle post repair and report to supervisor/service advisor if further inspection is required by another specialist	2	3	-	1
PC24. ensure completeness of tasks assigned before releasing the electric bicycle for the next procedure	-	1	-	1
PC25. dispose of materials such as old batteries, scrap of failed parts/aggregates as per organization's policies	1	2	-	1
PC26. return leftover consumable/parts, tools/equipment, and report if any malfunctions are observed to the person concerned	2	2	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N1457
NOS Name	Carry out routine service or minor repairs on electric bicycles and assist in diagnosis
Sector	Automotive
Sub-Sector	Automotive Vehicle Service
Occupation	Technical Service & Repair
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	28/07/2022
Next Review Date	28/07/2025
NSQC Clearance Date	28/07/2022

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).

Qualification Pack

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N9801.Organize work and resources (Service)	50	30	-	20	100	15
ASC/N1454.Assist in service, maintenance and repair of electric vehicle.	30	50	0	20	100	25
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	100	110	-	40	250	50

Elective: 1 Four Wheeler Electric Vehicle Service Assistant

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N1455.Automotive Detailer Technician	30	50	0	20	100	50
Total	30	50	-	20	100	50

Elective: 2 Two/Three wheeler Electric Vehicle Service Assistant

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N1456.Perform routine service, repair and maintenance of bicycle	30	50	-	20	100	50
Total	30	50	-	20	100	50

Elective: 3 Electric Truck/Bus Service Assistant

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N1457.Carry out routine service or minor repairs on electric bicycles and assist in diagnosis	30	50	-	20	100	50
Total	30	50	-	20	100	50

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
PwD	Persons with Disability
PwD	Persons with Disability
PwD	Persons with Disability

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.